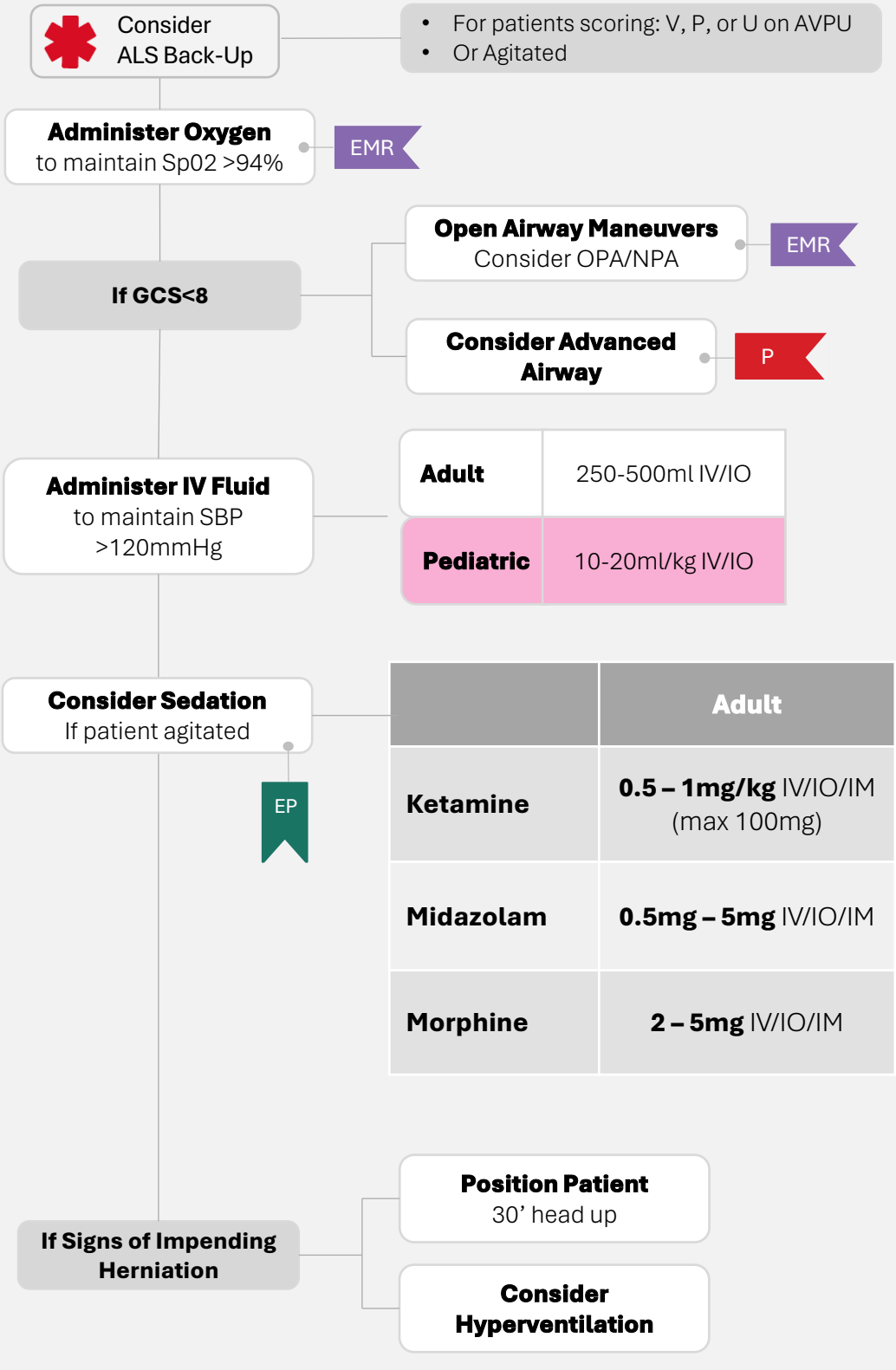

	Adult	Pediatric
Ketamine	**0.5 – 1mg/kg** IV/IO/IM (max 100mg)	**0.5 – 1mg/kg** IV/IO/IM, (max 100mg)
Midazolam	**0.5mg – 5mg** IV/IO/IM	**0.05mg/kg** IV/IO/IM
Morphine	**2 – 5mg** IV/IO/IM	**0.1mg/kg** IV/IO/IM, maximum 5mg

Patient Care Goals

1. Rapidly identify and manage traumatic brain injuries (TBI) to prevent secondary brain injury and optimize neurological outcomes.
2. Stabilize the patient's condition through targeted interventions, including oxygenation, ventilation and blood pressure management.
3. Ensure rapid transport to a trauma center with neurosurgical capabilities for definitive evaluation and treatment.

Patient Inclusion & Exclusion Criteria

Inclusion Criteria

- Patients with suspected TBI based on trauma history or clinical findings, including:
  - Blunt or penetrating head trauma (e.g., motor vehicle crashes, falls, assaults, or gunshot wounds).
  - Symptoms such as altered mental status, loss of consciousness, headache, vomiting, or neurological deficits (e.g., weakness, sensory loss).
  - Signs of severe TBI (e.g., unequal pupils, posturing, or seizures).
- Patients requiring assessment and management to prevent secondary brain injury (e.g., hypoxia, hypotension, or increased intracranial pressure).

Exclusion Criteria

- Patients with no history of head trauma or clinical findings suggestive of TBI (e.g., isolated extremity injuries without head involvement).
- Non-traumatic neurological conditions (e.g., stroke, hypoglycemia) mimicking TBI symptoms.

Key Patient Assessment Points

1. **Mechanism of Injury**
   - Identify the mechanism of head trauma (e.g., blunt vs. penetrating).
   - Confirm details such as height of fall, impact force, or presence of secondary injuries (e.g., facial fractures, cervical spine injury).
2. **Neurological Assessment**
   - Evaluate level of consciousness using AVPU (Alert, Verbal, Painful, Unresponsive) or Glasgow Coma Scale (GCS).
   - Assess for focal neurological deficits (e.g., unilateral weakness, sensory loss, or cranial nerve abnormalities).
   - Check pupil size and reactivity (e.g., unequal pupils, sluggish response) for signs of increased intracranial pressure (ICP).
3. **Signs of Severe TBI**
   - Note evidence of brain herniation (e.g., Cushing's triad: hypertension, bradycardia, irregular respirations).
   - Assess for posturing (decorticate or decerebrate), seizures, or cerebrospinal fluid (CSF) leakage from ears/nose.
4. **Trauma-Specific Findings**
   - Inspect for scalp lacerations, skull deformities, or open fractures indicating severe TBI.
   - Assess for associated injuries (e.g., cervical spine, chest, or abdominal trauma) that may contribute to hypoxia or hypotension.
5. **Determine Triage Category**
   - Determine Triage Category as per Patient Triage Categories CPG.
     - Red Priority:
       - GCS < or equal to 9, or progressive deterioration in GCS.

Patient Management & Interventions

1. **Consider ALS back-up**
   - Consider the requirement for ALS back-up as per Universal Care CPG.
   - ALS back-up should routinely be requested for:
     - Patients responding only to Pain or are Unresponsive, on AVPU scale.
2. **Primary Survey**

- Perform primary survey as per Universal Care CPG.
- Control catastrophic external bleeding.
- **Spinal Motion Restriction:**
    - The use of c-collars is contraindicated in patients with Traumatic Brain Injuries as it impedes cerebral venous damage, further increasing ICP.
    - Provide inline stabilization and support the head using the head block plate and head blocks – Forehead and Chin straps are not required.
    - Secure the patient to the stretcher.
- **Oxygenation:**
    - Titrate Oxygen to maintain SpO2 ≥94% to prevent hypoxia.
    - Secure airway with advanced techniques (e.g., endotracheal intubation) if GCS ≤8 or airway compromise.
    - Avoid hyperventilation; target EtCO2 35-45 mmHg to prevent cerebral vasoconstriction.

3. **Consider Red Flags**
    - GCS ≤8 or deteriorating neurological status indicating severe TBI or brain herniation.
    - Unequal pupils, posturing, or seizures suggesting increased ICP.
    - Hypotension (SBP <90 mmHg in adults) or hypoxia (SpO2 <90%), exacerbating secondary brain injury.
    - Signs of basilar skull fracture (e.g., Battle's sign, raccoon eyes, CSF otorrhea/rhinorrhea).
    - Associated polytrauma (e.g., chest or abdominal injuries) complicating TBI management.
4. **Secondary Survey**
    - Conduct a focused neurological and head examination to confirm TBI severity and monitor progression.
    - Obtain history of trauma mechanism and symptoms (e.g., loss of consciousness, vomiting) using OPQRST.
5. **Establish IV Access**
    - Indicated for fluid resuscitation, medication administration, or if deterioration is expected.
        - Consider intraosseous (IO) access if IV access cannot be obtained.
    - **Fluid Resuscitation**
        - Administer Normal Saline or Ringers Lactate for hypotension:
            - Adults: 500-1000 mL bolus IV/IO, reassess, target SBP ≥120 mmHg (or MAP ≥90 mmHg).
            - Avoid over-resuscitation to prevent cerebral edema.
6. **Advanced Airway Management**
    - If GCS <8:
        - Perform open airway maneuvers (Jaw thrust)
        - OPA/NPA may be used if the airway cannot be maintained with open airway maneuvers.
        - Consider Supraglottic Airway/ Intubation:
            - Advanced airway management in traumatic brain injury patients should be avoided (unless medication facilitated intubation is available).
            - Consider advanced airways if:
                1. The patient is profoundly unconscious (GCS=3, AND/
                2. The airway cannot be maintained with basic airway adjuncts (OPA and jaw thrust)
7. **Head Positioning**
    - If signs of impending herniation:
        - Elevate head of bed 30 degrees if no spinal precautions are needed to reduce ICP.
        - Maintain neutral head alignment to optimize cerebral venous drainage.
8. **Hyperventilation**
    - Note that it is very difficult to hyperventilate a patient who is not intubated and spontaneously breathing.
    - Hyperventilation should be initiated in a patient who has had hypoxemia and hypotension corrected, and has one (or more) of the following:
        - GCS score < 9 with extensor posturing (decerebrate posturing)
        - GCS score < 9 with asymmetric (or bilateral), dilated, or nonreactive pupils.
        - Initial GCS score < 9, who then drops his or her GCS by more than two points.
    - Ventilation should be adjusted to target an EtCO2 of 30-35mmHg.

- Pediatric ventilation rates should be adjusted based on the patient's age:

Suggested Rates for Hyperventilation in Traumatic Brain Injury

Age	Normal Ventilation Rate (ETC02 Target 35-45mmHg)	Hyperventilation Rate (ETCO2 Target 30-35)
Adults	12 breaths/min	20 breaths/min
Children	15 breaths/min	25 breaths/min
Infants	20 breaths/min	30 breaths/min

9. **Sedation**
   - Sedation should be administered to patients with agitation.
   - ALS back-up should be requested as soon as possible for agitated patients.
   - IV is the preferred method of administration for any sedation. Avoid IM administration whenever possible.
   - If IM administration is required, establish IVIO as soon as possible once patient is sedated.
   - Ketamine is the preferred sedation agent.
   - Reduced doses of Benzodiazepines or Morphine may also be administered if Ketamine is not available or contraindicated.
   - Dosage:
     - Ketamine:
       - Adults and Pediatric: 0.5-1mg/kg IV/IO/IM, maximum 100mg, repeat after 10 minutes if required.
     - Midazolam:
       - Adult: 0.5mg – 5mg IV/IO/IM, repeat after 5-10 minutes if required.
       - Pediatric: 0.05mg/kg IV/IO/IM, maximum 5mg, repeat after 10 minutes if required.
     - Morphine:
       - Adult: 2-5mg IV/IO/IM, repeat after 10 minutes if required.
       - Pediatric 0.1mg/kg IV/IO/IM, maximum 5mg, repeat after 10 minutes if required.
10. **Patient Reassessment**
   - Frequently Reassess the patient as per Universal Care CPG.
   - Specific Patient Reassessment Points:
     - GCS reassessment (Red Priority every 5 minutes, Yellow Priority every 10 minutes, Green Priority every 15 minutes).

Patient Safety Considerations

- Elderly Patients (>65 years)
  - Increased risk of TBI from low-energy mechanisms (e.g., ground-level falls) due to brain atrophy or anticoagulant use.
  - Monitor closely for delayed deterioration (e.g., subdural hematoma).
- Pediatric Patients (<13 years)
  - Pediatric TBI may present with subtle signs; prioritize transport for any neurological changes.
- ICP Management
  - Avoid hypoxia, hypotension, or hyperventilation, as these exacerbate secondary brain injury.
  - Maintain normocapnia (EtCO2 35-45 mmHg) if intubated to prevent cerebral vasoconstriction.

Clinical Insights and Additional Guidance

- None noted

Documentation

- Identification: Patient's full name, age, and gender

- Mechanism / Medical Complaint: Description of trauma mechanism (e.g., blunt or penetrating head injury, fall, assault, MVC) or reason for suspected TBI (e.g., witnessed loss of consciousness, altered mental status post-trauma).
- Injuries / Information: Findings from head and neurological assessment (e.g., visible skull fractures, contusions, pupillary response, GCS components), multisystem trauma evaluation, and indicators of increased ICP (e.g., cerebral edema or hemorrhage).
- Signs: Vital signs, ETCO2, presence of neurological deficits (e.g., posturing, lateralizing signs, Cushing's triad: hypertension, bradycardia, irregular breathing), pupillary findings (size, symmetry, reactivity), and signs of herniation (e.g., fixed dilated pupils).
- Treatment: All relevant treatment administered (e.g. Airway management, oxygen administration, IV fluid, sedation and tranexamic acid (TXA) if administered for suspected hemorrhage.
- Allergies: Any Known allergies.
- Medications: List any known medications the patient is taking.
- Background: Medical history affecting TBI management (e.g., anticoagulation use, prior TBI, coagulopathy, comorbidities like hypertension or diabetes influencing ICP risk).
- Other: Communication with receiving facility

Key Performance Indicators

1. Administer IV fluid for patients with a systolic blood pressure <120mmHg or MAP<90mmHg.

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